

VAISHALI VEERA SARAVANA PERUMAL

vaishaliveera04@gmail.com · +44 7887827977 · [linkedin.com/in/v-vaishali](https://www.linkedin.com/in/v-vaishali) · github.com/vaishali483

PROFESSIONAL SUMMARY

Explorer at heart, researcher in practice. I'm an MSc AI & ML student at the University of Birmingham, with hands-on experience in designing, training, and deploying machine learning systems across computer vision, robotics, and real-time pipelines. Co-author of 7 Scopus-indexed IEEE publications, I bring both research rigour and hands-on Python development to every project. My interests span LLMs, GenAI, Computer Vision, XR, robotics, and automation. I combine analytical rigour with design thinking to tackle problems that sit at the edge of what's possible. I am driven by curiosity and a commitment to building things that matter.

TECHNICAL SKILLS

Core Language	Python (primary · proficient) · C · C#
ML / DL	PyTorch · TensorFlow · Scikit-learn · Deep Learning · Reinforcement Learning · Neural Networks · Model Training, Fine-tuning & Evaluation · Predictive Analytics
ML Systems	Real-time Data Pipelines · Anomaly Detection · Digital Twin · Embedded Systems · Distributed Monitoring · Experimental Pipeline Design
Computer Vision	OpenCV · YOLO · MediaPipe · Image Segmentation & Recognition · Image Processing
NLP / GenAI	LLMs · Generative AI · RAG · Prompt Engineering · NLP
Data Science & Analysis	Data Visualisation · Tableau · MongoDB · MySQL · Firebase
Backend & APIs	Fast API · REST APIs
Research Methods	Empirical Validation · Model Benchmarking · Experimental Design · Scientific Writing
XR & Spatial	Unity 3D · AR/VR · Spatial Computing · Blender · Autodesk Fusion 360
Other Tools	Adobe Photoshop · Figma · Agile Project Management · Sensor Fusion · Embedded Systems

EDUCATION

MSc Artificial Intelligence and Machine Learning <i>University of Birmingham, United Kingdom</i>	Obtained 75.67%	2025 – 2026
B.Tech Artificial Intelligence and Data Science <i>Kumaraguru College of Technology, India</i>	Obtained 86.6%	2021 – 2025

WORK EXPERIENCE

Research Associate <i>Extreme Robotics Lab, University of Birmingham</i>	Dec 2025 – Present
- Design and execute end-to-end empirical ML pipelines to benchmark SOTA Computer Vision and deep learning architectures against robotic perception tasks, generating reproducible experimental results aligned with academic publication standards. - Analyse model training dynamics, diagnose performance bottlenecks, and iterate on deep learning architectures — directly mirroring the model debugging and evaluation work central to the ML Systems & Performance workstream.	
Unity Developer Intern (AR/AI) <i>Schnell Energy Equipments</i>	Aug 2024 – Mar 2025
- Fine-tuned and deployed deep learning image segmentation models within a production AR pipeline (Unity 3D + ARCore), integrating ML inference into a real-time, operational system used for smart city lighting maintenance. - Analysed model performance under real-world deployment constraints and iterated on architecture choices to meet operational reliability requirements — demonstrating the research-to-engineering balance the role demands. - Engineered ML inference pipelines bridging Python-based deep learning models and live AR environments, building practical experience with the full ML systems stack from training through deployment.	
Graduate Innovation Engineer <i>Forge Innovation & Ventures</i>	Jan 2024 – Jul 2024
- Developed ML-driven prototypes spanning Computer Vision, NLP, and Industrial Automation, applying engineering rigour to translate research findings into deployable, reliable systems within tight project cycles. - Worked cross-functionally alongside engineers, hardware specialists, and domain experts — building the collaborative, multi-disciplinary instincts required for research-engineering environments like Anthropic.	

- ▶ Gained hands-on exposure to ROS and Computational Hardware, developing systems-level thinking about how ML models interact with physical infrastructure and real-world constraints.

Unity Developer Intern | [SatoriXR, IITM Research Park](#)

Jul 2023 – Aug 2023

- ▶ Built functional AR applications using Unity 3D and C#, developing production software engineering discipline and an appreciation for reliability in deployed interactive systems.

KEY PROJECTS

Open-Set Kitchen Hazard Detection: VLM-Grounded LLM Agent with Unity AR (MSc Dissertation)

- ▶ Built an end-to-end kitchen safety system combining YOLO11s, Google Gemini, FastAPI, and Unity AR to detect and explain visible hazards from sampled camera frames, with present/possible risk classification and validated exact, approximate, or warning-only localisation.
- ▶ Implemented cooking-session tracking with periodic frame analysis, lightweight textual context, deduplicated session summaries, local session storage, and an Android AR interface for real-time safety warnings.

Digital Twin for Real-Time 3D Printer Monitoring and Anomaly Detection

- ▶ Architected a real-time data pipeline using digital twin technology to ingest live sensor streams, simulate printer state, and visualise operational parameters on a unified dashboard — directly analogous to the on-demand infrastructure and data pipeline projects in this workstream.
- ▶ Built an end-to-end ML anomaly detection pipeline integrating sensor data ingestion, fault classification, and automated alerting, demonstrating experience constructing complex, data-intensive systems that balance research innovation with operational reliability.
- ▶ Designed infrastructure supporting remote parameter control and real-time state synchronisation across distributed components, gaining practical experience with monitoring and debugging live ML-integrated systems.

AI Keyword Generator for SEO

- ▶ Built a GenAI-powered SEO keyword research app using Python, Streamlit, Google Gemini, Pandas, Pydantic, and scikit-learn to generate, classify, cluster, and prioritise keywords from user-provided topics.
- ▶ Combined LLM-based search intent classification with TF-IDF and KMeans clustering to group related keywords and generate explainable SEO insights and content recommendations.
- ▶ Developed an interactive Streamlit dashboard with filtering, visualisations, CSV/Markdown export, error handling, and automated Pytest/AppTest coverage using mocked Gemini responses.

ML Model for Production Parameter Prediction Manufacturing Industries

- ▶ Trained, evaluated, and iteratively improved a machine learning model to predict optimal manufacturing parameters — applying feature engineering, cross-validation, and hyperparameter tuning to achieve production-quality predictive accuracy.
- ▶ Analysed training dynamics and model behaviour across dataset conditions, building systematic debugging intuition for ML model performance.

Gesture-Controlled 6 DoF Robotic Arm via Inverse Kinematics and Deep Learning

- ▶ Implemented real-time gesture recognition using MediaPipe deep learning models and custom inverse kinematics algorithms, deploying the full pipeline on embedded hardware to enable precise human-robot interaction.
- ▶ Integrated ML inference with physical actuator control, gaining end-to-end systems experience from model training and evaluation through to low-latency deployment on constrained hardware.

RESEARCH PUBLICATIONS

- ▶ **Scopus / Springer Journal (Accepted)** · Digital Twin based Real-Time Anomaly Detection for FDM Additive Manufacturing
- ▶ **Scopus / IEEE Xplore (Jun 2025)** · [AI Powered Monitoring and Risk Prediction for Maternal Health to Ensure Fetal Well-Being](#)
- ▶ **Scopus / IEEE Xplore (Jun 2025)** · [Sustainable Urban Street Lighting through Predictive Maintenance using IoT and AI](#)
- ▶ **Scopus / IEEE Xplore (Feb 2024)** · [Implementing Catboost Algorithm for Allergen Cross Contamination Detection in Food Industry](#)
- ▶ **Wiley Scrivener Publishing (Feb 2024)** · [A Novel Adaptive Framework for Immersive Learning Using VR in Education](#)
- ▶ **IGI Global (Jan 2024)** · [Integration of UAV Systems with ML Algorithms for Wildlife Monitoring and Conservation](#)
- ▶ **Scopus / IEEE Xplore (Aug 2023)** · [Collaboration of Mixed Reality for Interactive Visualisation of Ocean Mapping](#)
- ▶ **Scopus / IEEE Xplore (Aug 2023)** · [Industry 5.0: Enhancing Human-Robot Collaboration through Collaborative Robots — A Review](#)

VOLUNTEERING

- ▶ Student Enterprise Ambassador, University of Birmingham — supporting entrepreneurship and innovation initiatives, and building community across the student research and engineering ecosystem.